

Arhaan Penwala

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering (AI and Robotics), CGPA: **9.13/10.0**

Chennai, India

Expected May 2027

EXPERIENCE

Co-Founder and CTO

Nov '24 – Present

V Help | Flutter, Supabase (PostgreSQL, Deno Edge Functions), Firebase, Azure

VIT Chennai

- Own backend architecture for a live campus super-app serving **1,000+ registered students**, as top contributor of 3 core engineers
- Closed a privilege-escalation surface by relocating **24** client write paths into server-authorized Deno Edge Functions backed by PostgreSQL Row Level Security
- Migrated **500+ accounts** from two separate Firebase projects onto a single Supabase identity via a custom JWT exchange, at **zero re-registration**
- Cut Android release turnaround from **45 minutes** of manual work to **6 minutes** by automating APK build and distribution through GitHub Actions CI/CD

Software Development Engineer Intern

Dec '25 – Feb '26

BPO Integra India Private Limited

Remote

- Cut response times **23%** across **4** production REST endpoints by collapsing N+1 query patterns and adding composite MySQL indexes
- Shipped Node.js/Express REST services and a reusable React component library across **6** pages of a live production site

AI Research Intern

May – Jun '25

The Indian Hotels Company Limited (Taj Hotels, Tata Group)

Mumbai, India

- Built a Python cost model over **5** enterprise LLM vendors and **15** hospitality use cases, projecting a **35%** spend reduction that drove leadership's vendor selection

PROJECTS

PolarisGCS | FastAPI, PostgreSQL/TimescaleDB, Redis, React/TypeScript, Docker, Kotlin

Feb '26 – Present

- Ingested telemetry from **15 concurrent MAVLink vehicles** under ArduPilot SITL into stateless FastAPI instances, fanning out over Redis pub/sub so servers scale horizontally with no in-process state
- Persisted time-series telemetry in TimescaleDB with Alembic migrations and continuous aggregates, bounding storage growth while retaining full historical playback
- Containerized FastAPI, PostgreSQL, Redis, and object storage into a **single-command** Docker Compose stack, hardening auth with hashed device tokens and **role-gated** admin APIs

Queez | FastAPI, MongoDB, Redis, WebSockets, Socket.IO, Docker, Flutter

Jun '25 – Jan '26

- Eliminated a scoring **race condition** in real-time multiplayer sessions using per-user and per-session Redis distributed locks with retry and backoff, as author of **51 of 52** backend commits
- Validated concurrency under load with a Python harness spawning **50 simultaneous bot clients** driving randomized gameplay against live WebSocket sessions
- Removed a MongoDB round trip on every answer submission by caching active quiz questions behind a **1-hour TTL** Redis read-through layer
- Blocked client-side score manipulation with server-side timestamp validation, sub-0.5s answer flooring, and a maximum-score ceiling against overflow exploits

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, Kotlin, Dart

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.IO, Redis Pub/Sub, JWT Authentication, Deno Edge Functions

Databases: PostgreSQL/TimescaleDB, MySQL, MongoDB, Redis, SQLite, Supabase, Firebase

Frontend & Infrastructure: React.js, TypeScript, Vite, Flutter, Riverpod, PySide6, Docker, AWS, Azure, Linux, Git, GitHub Actions, Postman

Concepts: Data Structures and Algorithms, Object Oriented Programming, System Design, REST API Design, Concurrency, Caching, Database Indexing, CI/CD

ACHIEVEMENTS AND PUBLICATIONS

2nd Place out of **500+ teams**, Hack N Droid '25 | **Publication:** gait dataset, IEEE DataPort, DOI 10.21227/ng5-b637